

MOHAMMAD ADIL KHAN

+91-8528005726 | adilkhan.mohd20@gmail.com | github.com/adilkhan-m20 | linkedin.com/in/adil-khan20 | leetcode.com/adilkhan_m20

EDUCATION

Ramaiah Institute of Technology

Bachelor's of Engineering in Computer Science – CGPA: 9.25

Boys High School & College

Senior Secondary Education (XII) – Percentage: 93.4%

Boys High School & College

Secondary Education (X) – Percentage: 92%

Oct 2023 – Present

Bengaluru, India

April 2021 – March 2022

Prayagraj, India

April 2019 – March 2020

Prayagraj, India

COURSEWORK / SKILLS

- DSA
- OOPS Concepts
- DBMS
- Operating Systems
- Computer Networks
- Software Engineering

EXPERIENCE

Samsung R&D Institute, Bengaluru

Project Intern

Dec 2025 – Present

Bengaluru, India

- Designed and implemented a **semantic search framework** for SmartThings to enable natural language-based device discovery across heterogeneous smart home catalogs.
- Developed a **text embedding and vector search pipeline** to capture semantic relationships between device capabilities, attributes, and user queries beyond keyword-based search.
- Built and exposed **search APIs**, and evaluated system performance using domain-specific metadata and sample user queries to ensure accurate and relevant retrieval.

PROJECTS

Natural Language to SQL Conversion | FAISS, Sentence-Transformers, SQLite, Ollama, Vector Databases (Chroma)

- Developed a Retrieval-Augmented Generation (**RAG**) pipeline for natural language to SQL conversion, leveraging semantic search and schema-aware context injection.
- Implemented **vector embeddings** and **FAISS-based similarity** search, indexing structured database schema chunks
- Integrated a **local Large Language Model (LLM)** for **offline SQL generation**, improving data privacy, reducing inference latency, and eliminating external API dependencies.

Deepfake Immunization Framework | PyTorch, HuggingFace Diffusers, CLIP, OpenCV, FastAPI

- Designed a 7-phase adversarial machine learning pipeline to proactively protect digital identities by injecting **mathematically bounded noise (PGD)** into image latent spaces, achieving complete image immunization in **30–40 seconds**.
- Engineered an **intent-aware optimization** engine utilizing **CLIP text embeddings** and **Expectation Over Transformation (EOT)** to neutralize malicious generative attacks while preserving high image utility (**>95% SSIM, ~45 dB PSNR**).
- Developed an automated **Red-Team evaluation suite** to simulate real-world **ControlNet** and **Stable Diffusion attacks**, completing comprehensive defense validation in **70–80 seconds**.
- Quantified defense success using a custom **Deepfake Resistance Score (DRS)** and **ArcFace identity verification (distance > 0.68)**, successfully blocking around **80–100%** of tested malicious identity alterations across a distributed dual-GPU environment.

Research Paper – “Comparative Analysis of Regression Models for Predicting Superconducting Critical Temperature”

Accepted and Presented at Health Informatics, Intelligent Systems and Networking Technologies (HINT '25)

- Conducted a comparative analysis of **six advanced regression models** to predict superconducting critical temperature (T_c).
- Leveraged a large dataset of over **21,000 superconductors** to train and evaluate the models.
- Achieved a high predictive accuracy with an **RMSE of 9.02K** and an **R^2 score of 0.93**, demonstrating the effectiveness of ML in accelerating materials discovery.

TECHNICAL SKILLS

Languages: Java, C, JavaScript, SQL, MongoDB, HTML, CSS

Machine Learning & AI: Supervised & Unsupervised Learning, Regression, Classification, Clustering, Reinforcement Learning, RAG, Model Evaluation, Deep Learning

Frameworks/Libraries: Express, NodeJS, React, Tensorflow, Numpy, Pandas, LangGraph, LangChain, scikit-learn

CERTIFICATIONS / COURSES

- Deep Learning Specialization - *DeepLearning.AI*
- Software Engineering - *NPTEL (IIT Kharagpur)*
- Cloud Computing - *NPTEL (IIT Kharagpur)*
- Go for Gold, Gold Level Certification - *Accenture iAspire (ATCI)*